From microgravity boundary-layer biophysics to the spaceflight multi-ome: a coupled lattice-Boltzmann and Farquhar–von Caemmerer–Berry model of *Arabidopsis* gas exchange and enclosure carbon starvation

Richard J. Barker[a,*]

[a] *Purdue University, West Lafayette, IN, USA*

Abstract

Plants grown in spaceflight display conserved transcriptional and proteomic stress signatures whose physical origin remains debated. Two distinct gas-transport mechanisms have been proposed: the microgravity loss of buoyancy-driven free convection, which thickens the unstirred diffusive boundary layer around leaves, and atmospheric stagnation inside sealed or poorly ventilated spaceflight growth hardware. Because no flight experiment has resolved sub-millimetre leaf boundary layers or logged in-canister CO_2 across hardware classes, we built an end-to-end biophysical and multi-omics framework to test both mechanisms quantitatively. First, we implemented a two-dimensional D2Q9/D2Q5 lattice-Boltzmann computational fluid dynamics (CFD) solver for incompressible flow, multi-species (CO_2, O_2, H_2O) advection–diffusion, and solutal Boussinesq buoyancy under an adjustable gravity vector ($1\,g \rightarrow$ Mars $0.38\,g \rightarrow$ Moon $0.17\,g \rightarrow \mu g$), validated against four numerical benchmarks (lid-driven cavity, Kármán cylinder wake, natural-convection cavity to 0.18% on Nusselt number, and analytic erfc diffusion) and anchored to empirical *Arabidopsis thaliana* whole-chamber gas exchange ($A = 3.85\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$). Removing gravity eliminates convective plumes ($u_{\max} \propto \sqrt{g} \rightarrow 0$), reducing leaf boundary-layer conductance g_{bl} from 0.997 to $0.443\,\mathrm{mol\,m^{-2}\,s^{-1}}$ for an isolated leaf, 0.540 to $0.272\,\mathrm{mol\,m^{-2}\,s^{-1}}$ for a rosette, and 0.133 to $0.058\,\mathrm{mol\,m^{-2}\,s^{-1}}$ for a dense microgreen canopy. Forced ventilation reverses this microgravity penalty, but the Earth-equivalent airflow scales with planting density from $\approx 2.0\,\mathrm{cm\,s^{-1}}$ (leaf) to $\approx 6.1\,\mathrm{cm\,s^{-1}}$ (rosette). Second, we coupled the CFD-derived g_{bl} values and zero-dimensional (0-D) hardware mass balances for three International Space Station growth systems—sealed Biological Research in Canisters (BRIC), micropore-taped Characterizing Arabidopsis Root Attractions (CARA), and actively ventilated Vegetable Production System (VEGGIE)—to a Farquhar–von Caemmerer–Berry (FvCB) C_3 model with explicit Rubisco oxygenation ($\varphi = V_o/(V_o + V_c)$). An illuminated sealed BRIC canister draws its headspace CO_2 down to the compensation point within $\sim 7\,\mathrm{min}$, collapsing 12 h carbon gain to $\approx 1\%$ of an unenclosed control (versus $\approx 90\%$ for CARA and $\approx 100\%$ for VEGGIE) and raising φ from 18.0% to 46.2% at experimental irradiance ($60\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$). Because canister drawdown is a mass balance, it occurs symmetrically in both flight and ground controls: it cancels out of the published flight-versus-ground contrast while shifting the baseline metabolic state into severe carbon starvation (an enclosure effect $\approx 22\times$ larger than the microgravity boundary-layer effect). Furthermore, contrary to our starting hypothesis, CO_2 exhaustion causes the $1\,g$ and μg boundary layers to converge rather than diverge as photosynthetic flux vanishes, leaving a residual microgravity assimilation deficit of $< 3\%$ at leaf/rosette scale and up to 11% in a dense canopy at $60\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ (5–9% and 26–29% at saturating light). Third, after an exhaustive census of all 567 studies in the NASA Open Science Data Repository (OSDR) revealed zero plant metabolomics datasets (0 of 12 metabolomics studies), we formulated a falsifiable

*Corresponding author. ORCID: 0000-0001-5681-9857.

Email address: dr.richard.barker@gmail.com (Richard J. Barker)

21-compound flux-propagated metabolite scaffold and tested the coupled model against OSD-522 (BRIC-LED) shoot transcriptomics and proteomics and across six matched flight-versus-ground contrasts from five OSDR studies. In OSD-522, the model called eight of nine transcriptome gene-set responses and four of seven in the proteome (repression of photosystems, Rubisco, carbon fixation, and starch/sucrose metabolism; induction of *DIN6/ASN1* carbon-starvation markers; maintenance of photorespiratory enzymes above Rubisco). Multi-omics pathway integration in PaintOmics 4 showed that endoplasmic reticulum protein processing (unfolded protein response, $p = 8.6 \times 10^{-9}$) dominates the global response while photorespiration ranks in the bottom 8% (214th of 232 pathways). Across the six OSDR contrasts, illumination separated every lit study from every dark study on carbon starvation over photosynthesis ($p = 0.05$), whereas enclosure class alone did not. We conclude that unlogged enclosure atmosphere is a first-order covariate in spaceflight plant biology and outline the in-canister sensor logging, 20-compound targeted metabolomics panel, and three-arm / 2×2 factorial experimental designs required to separate hardware carbon starvation from microgravity.

Keywords: microgravity, boundary layer, computational fluid dynamics, lattice-Boltzmann, Farquhar–von Caemmerer–Berry, photorespiration, *Arabidopsis thaliana*, spaceflight growth hardware, multi-omics, NASA OSDR

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1. Introduction

On Earth, a photosynthesising or transpiring leaf warms and humidifies the thin layer of air in contact with its epidermis. Because water vapour ($M = 18\,\mathrm{g\,mol^{-1}}$) is lighter than dry air ($M \approx 29\,\mathrm{g\,mol^{-1}}$), the humidified boundary layer is buoyant and rises in a free-convection plume that continuously sweeps the leaf surface, replenishing CO_2 while removing O_2 and water vapour. This buoyancy-driven ventilation is invisible under terrestrial conditions, yet it sets the still-air boundary-layer conductance (g_{bl}) that acts in series with stomatal (g_s) and mesophyll (g_m) conductances to govern the chloroplastic CO_2 mole fraction (C_c) reaching Rubisco:

$$C_a \xrightarrow{g_{bl}} C_s \xrightarrow{g_s} C_i \xrightarrow{g_m} C_c, \tag{1}$$

where C_a, C_s, and C_i denote the bulk enclosure, leaf-surface, and intercellular CO_2 mole fractions, respectively.

In microgravity (μg), hydrostatic pressure gradients and buoyancy-driven natural convection vanish. In the absence of forced ventilation, gas transport outside the organ collapses to pure molecular diffusion, thickening the unstirred boundary layer (δ) and steepening the surface-to-bulk concentration gaps (ΔC) of CO_2, O_2, and H_2O [1, 2]. For decades, this boundary-layer thickening has been invoked to explain the hypoxia, oxidative stress, altered carbon metabolism, and morphological phenotypes observed in spaceflight-grown plants [1, 3, 4], and it forms the primary engineering rationale for active fan ventilation in modern orbital growth chambers. At the biochemical level, Rubisco partitions between carboxylation (V_c) and oxygenation (V_o) in direct proportion to the ratio of O_2 to CO_2 at the catalytic site [5]:

$$\frac{V_o}{V_c} = \frac{2\Gamma^*}{C_c}, \qquad \varphi = \frac{V_o}{V_o + V_c} = \frac{2\Gamma^*}{C_c + 2\Gamma^*}, \tag{2}$$

where Γ^* is the photorespiratory CO_2 compensation point in the absence of day respiration (R_d). Any transport limitation that depresses C_c or elevates local O_2 shifts Rubisco toward oxygenation, accelerating photorespiration ($R_p = 0.5\,V_o$) at the expense of net carbon assimilation ($A = V_c - 0.5\,V_o - R_d$).

Interpreting molecular data from spaceflight experiments, however, requires confronting a second, larger transport barrier: the growth enclosure itself. Plants flown on the International Space Station (ISS) have been cultivated across three distinct hardware classes that differ by orders of magnitude in gas exchange with the spacecraft cabin:

1. **Hermetically sealed canisters**, exemplified by Biological Research in Canisters (BRIC) and BRIC-LED Petri Dish Fixation Units [3, 6];
2. **Passively vented Petri dishes**, exemplified by Characterizing Arabidopsis Root Attractions (CARA), where 120 mm square plates are sealed with gas-permeable 3M Micropore surgical tape and mounted against the cabin wall [7, 8]; and
3. **Actively ventilated growth chambers**, exemplified by the Vegetable Production System (VEGGIE) and Advanced Plant Habitat (APH), which circulate cabin air across the stand with fans [9].

Ground studies have demonstrated that sealing Petri dishes with impermeable films (Parafilm or polyethylene wrap) rapidly depletes headspace CO_2 in the light and induces thousands of stress-response genes compared with gas-permeable surgical tape [10, 11]. Similarly, a meta-analysis of 15 *Arabidopsis* spaceflight transcriptomes identified growth hardware and illumination as the largest confounding variables across the archive [4]. Yet to our knowledge, no BRIC-class spaceflight experiment has ever logged the internal CO_2 and O_2 concentrations experienced by its plants, and no whole-chamber sensor resolves the sub-millimetre leaf boundary layer.

Here we bridge fluid mechanics, photosynthetic biochemistry, and spaceflight multi-omics in a single reproducible framework. We first construct and validate an in-browser two-dimensional lattice-Boltzmann CFD solver (`LunarLeaf-CFD`) that resolves incompressible flow, multi-species ($CO_2/O_2/H_2O$) advection–diffusion, and solutal Boussinesq buoyancy across gravity regimes ($1\,g$, Mars $0.38\,g$, Moon $0.17\,g$, and μg), three plant architectures (single leaf, overlapping rosette, and dense microgreen canopy), forced-ventilation sweeps, and the three ISS hardware boundary conditions (BRIC, CARA, VEGGIE). We then pass the CFD-derived boundary-layer conductances (g_{bl}) and 0-D enclosure mass balances into an FvCB C_3 carboxylation/oxygenation model, propagate the resulting metabolic fluxes into a falsifiable 21-compound metabolite scaffold, and test the model's predictions against NASA Open Science Data Repository (OSDR) transcriptomics and proteomics from OSD-522 (BRIC-LED) [6] and across six matched flight-versus-ground contrasts spanning five independent spaceflight experiments [6, 8, 9, 12, 13].

2. Results

2.1. Numerical verification and empirical gas-exchange anchoring of the lattice-Boltzmann solver

Before modelling biological geometries, the two-dimensional incompressible-flow (D2Q9 Bhatnagar–Gross–Krook, BGK) and three-species advection–diffusion (D2Q5) lattice-Boltzmann solver was verified against four independent numerical benchmarks with exact or established reference solutions (Table 1A). First, in the lid-driven square cavity at Reynolds number $\mathrm{Re} = 100$, the steady centreline velocity profiles $(u_x(y), u_y(x))$ matched the multigrid benchmark of Ghia et al. [14] to a relative L_2 error of 0.0105 (gate threshold $L_2 < 0.05$). Second, for laminar flow past a circular cylinder at $\mathrm{Re} = 100$ in a channel of blockage ratio $D/H = 0.17$, the Kármán vortex-shedding Strouhal number measured over 16 shedding cycles was $\mathrm{St} = 0.192$, matching the expected confined-channel regime ($\mathrm{St} \approx 0.19$–0.20; unconfined reference $\mathrm{St} = 0.164$). Third, in a differentially heated square cavity at Rayleigh number $\mathrm{Ra} = 10^4$ and Prandtl number $\mathrm{Pr} = 0.71$, the hot-wall Nusselt number converged to $\mathrm{Nu} = 2.242$ against the de Vahl Davis [15] benchmark of $\mathrm{Nu} = 2.238$, an error of 0.18%. Fourth, one-dimensional step-change scalar diffusion reproduced the analytic $\frac{1}{2}\mathrm{erfc}(x/\sqrt{4Dt})$ profile

Table 1: **Lattice-Boltzmann solver verification gates and empirical *Arabidopsis* gas-exchange calibration.** **A**, Headless numerical benchmark gates (`validation/gates.ts`). **B**, Empirical whole-plant chamber gas-exchange anchor [18, 19] and lattice-to-physical unit mapping (`T1_measured_gas_exchange.csv`, `T3_calibration.csv`).

A. Numerical validation benchmarks

Gate	Benchmark problem	Reference solution	LBM (BGK / MRT)	Status
1	Lid-driven cavity ($\mathrm{Re} = 100$)	Ghia et al. [14] centreline ($L_2 < 0.05$)	$L_2 = 0.0105$ / 0.0115	Pass
2	Cylinder wake ($\mathrm{Re} = 100$, confined)	Kármán shedding ($\mathrm{St} \approx 0.19$–0.20)	$\mathrm{St} = 0.192$ / 0.200	Pass
3	Natural convection ($\mathrm{Ra} = 10^4$, $\mathrm{Pr} = 0.71$)	de Vahl Davis [15] ($\mathrm{Nu} = 2.238$)	$\mathrm{Nu} = 2.242$ (0.18%) / 2.248	Pass
4	Pure 1-D diffusion	Analytic $\frac{1}{2}\mathrm{erfc}(x/\sqrt{4Dt})$	$L_2 < 10^{-4}$ / 10^{-4}	Pass

B. Empirical gas-exchange anchor and physical lattice calibration

Parameter	Physical quantity	Value	Units	Source / Derivation
A_{net}	Net CO_2 assimilation (Col-0 WT)	3.85 (1.38)	$\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ ($\mu\mathrm{mol\,h^{-1}\,cm^{-2}}$)	Whole-plant chamber [18, 19]
R_d	Dark respiration ($R/A = 0.31$)	1.18 (0.43)	$\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ ($\mu\mathrm{mol\,h^{-1}\,cm^{-2}}$)	Whole-plant chamber [18, 19]
ΔO_2	Closed-chamber diel O_2 range	6.6–32.8	% (over 4.8 d)	Vernier closed-chamber trace
L_{leaf}	Characteristic leaf chord (52 cells)	1.50	cm	*Arabidopsis* blade
Δx, Δt	Lattice spacing and time step	0.288, 0.173	mm, ms	Matched to $D_{H_2O} = 2.4 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$
u^*	Lattice-to-physical velocity scale	1.66	$\mathrm{m\,s^{-1}}$ / lattice unit	$\Delta x/\Delta t$ (1 g leaf $u_{\max} = 7.7\,\mathrm{cm\,s^{-1}}$)
D_k	Diffusivities ($D_{CO_2} : D_{O_2} : D_{H_2O}$)	1.6 : 2.0 : 2.4	$\times 10^{-5}\,\mathrm{m^2\,s^{-1}}$	Air at 25 °C

to $L_2 < 10^{-4}$. An opt-in multiple-relaxation-time (MRT) collision operator [16] passed the same four gates ($L_2 = 0.0115$, $\mathrm{St} = 0.200$, $\mathrm{Nu} = 2.248$, erfc $L_2 = 0.0001$). Gate 3 is the decisive physical test: reproducing the natural-convection benchmark to 0.18% confirms that the Guo body-force coupling [17] between scalar density perturbations and gravity is quantitatively accurate, ensuring that scaling $|\mathbf{g}| \to 0$ genuinely collapses convective transport to the pure diffusion limit certified by Gate 4.

To anchor the surface source/sink terms in physical units, we used two independent whole-chamber gas-exchange datasets for wild-type *Arabidopsis thaliana* Col-0 (Table 1B; Fig. 1a,b). Whole-plant chamber measurements from the Framework Model v2 dataset [18, 19] establish a mean net CO_2 assimilation rate of $A = 1.38\,\mu\mathrm{mol\,h^{-1}\,cm^{-2}} = 3.85\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ and a dark respiration rate of $R = 0.43\,\mu\mathrm{mol\,h^{-1}\,cm^{-2}} = 1.18\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ ($R/A = 0.31$). An independent 4.8 d closed-chamber Vernier sensor trace (Fig. 1a; Supplementary Fig. S1) confirmed strong plant-driven diel O_2 cycling (6.6–32.8%) and humidity oscillations. Resolving a 1.5 cm leaf blade across 52 lattice cells fixes the spatial resolution at $\Delta x = 0.288$ mm and time step at $\Delta t = 0.173$ ms, yielding a free-convection plume velocity of $u_{\max} \approx 7.7\,\mathrm{cm\,s^{-1}}$ above an Earth-gravity leaf. When operated as a hermetically sealed chamber with a central stomatal source, the solver reproduced linear gas accumulation while conserving total species mass to a relative error of 2×10^{-5} (Fig. 1c) and established a persistent concentration gap between the near-leaf surface and the far chamber wall (Fig. 1d).

2.2. Gravity and plant architecture control the diffusive boundary layer

We next evaluated steady-state gas transport (150,000 steps ≈ 26 s, or 900,000 steps ≈ 156 s for the slowly diffusing μg canopy; drift 0.00% over the final 5 s in all cases; Supplementary Fig. S2a and Table S3) across four gravity levels—Earth ($1\,g = 9.81\,\mathrm{m\,s^{-2}}$), Mars ($0.38\,g$), Moon ($0.17\,g$), and microgravity ($0\,g$)—and three geometric scales: an isolated leaf blade, an overlapping eight-leaf rosette, and a dense microgreen canopy of upright shoots spaced 3.5 mm apart (Fig. 2, Fig. 3, Table 2).

At $1\,g$, humid, CO_2-depleted air rises off the leaf in a buoyant plume ($\mathrm{Ra}_{H_2O} = 1.4 \times 10^4$), sweeping the upper blade surface and maintaining an effective boundary-layer conductance of $g_{bl} = 0.997\,\mathrm{mol\,m^{-2}\,s^{-1}}$ and an equivalent diffusive film thickness of $\delta = D_{CO_2}/g_{bl} = 0.67$ mm ($\mathrm{Sh} = 22.4$; Table 2). Reducing gravity through Martian and Lunar levels weakens convective velocity as $u_{\max} \propto \sqrt{g}$ ($4.61 \times 10^{-2} \to 2.75 \times 10^{-2} \to 1.65 \times 10^{-2} \to 0$ lattice units), causing δ to thicken monotonically to 0.79 mm (Mars), 0.93 mm (Moon), and 1.50 mm in microgravity ($g_{bl} = 0.443\,\mathrm{mol\,m^{-2}\,s^{-1}}$, a 2.25× reduction; Fig. 3 left).

Plant architecture compounds the microgravity transport penalty (Fig. 3 right; Table 2). In an overlapping rosette, leaves shelter the central crown, reducing g_{bl} from $0.540\,\mathrm{mol\,m^{-2}\,s^{-1}}$ on Earth to $0.272\,\mathrm{mol\,m^{-2}\,s^{-1}}$ in μg ($\delta = 2.45$ mm) and elevating the peak crown CO_2 drawdown to 25.4 ppm

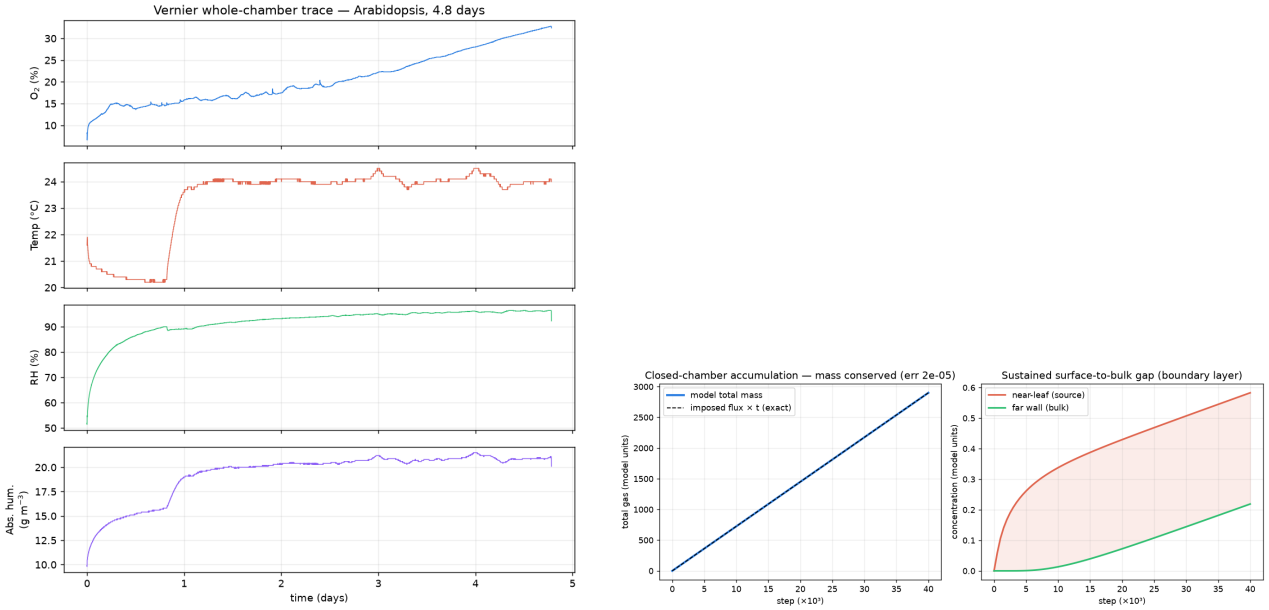


Figure 1: **Empirical gas-exchange anchoring and closed-chamber transport validation. Left**, Measured 4.8 d closed-chamber Vernier trace of O_2 (%), temperature (°C), relative humidity (%), and absolute humidity ($g\,m^{-3}$) around *Arabidopsis thaliana*, demonstrating diel photosynthetic/respiratory gas cycling (`F1_vernier_timeseries.png`). **Right**, Closed-chamber lattice-Boltzmann validation (`F5_chamber_validation.png`): total chamber gas tracks the integrated stomatal flux to a relative mass-conservation error of 2×10^{-5}, while a sustained surface-to-bulk concentration gap develops between the near-leaf fluid and the far wall.

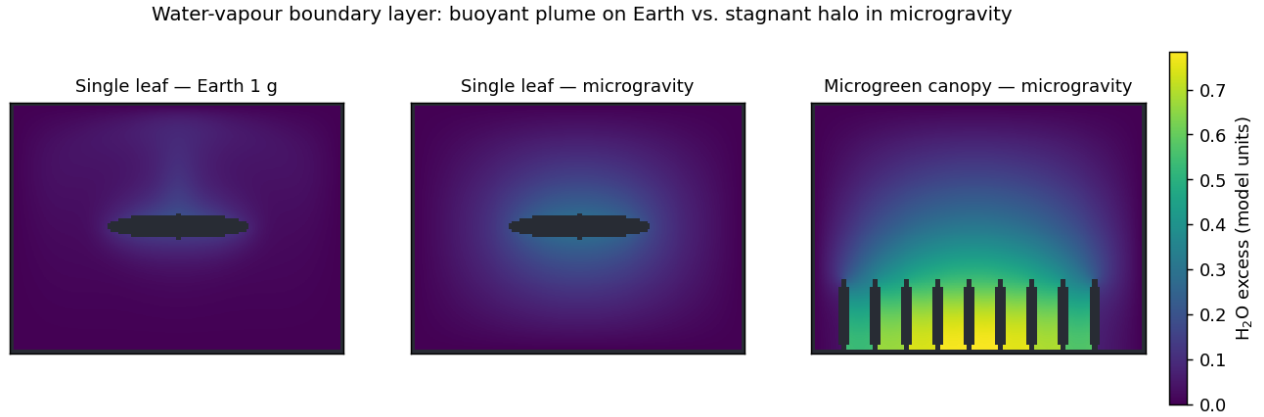


Figure 2: **Steady-state water-vapour boundary layers across gravity and plant architecture.** Simulated H_2O concentration excess (model units) at steady state (`F3_plume_maps.png`). At $1\,g$ (**left**), solutal buoyancy drives an upward convective plume that thins the upper-surface boundary layer. In microgravity (**centre**), buoyant convection vanishes ($u_{\max} = 0$) and a thick, symmetric diffusive halo surrounds the isolated leaf. In a microgravity microgreen canopy (**right**), overlapping diffusive halos merge between adjacent shoots, trapping stagnant air throughout the stand interior.

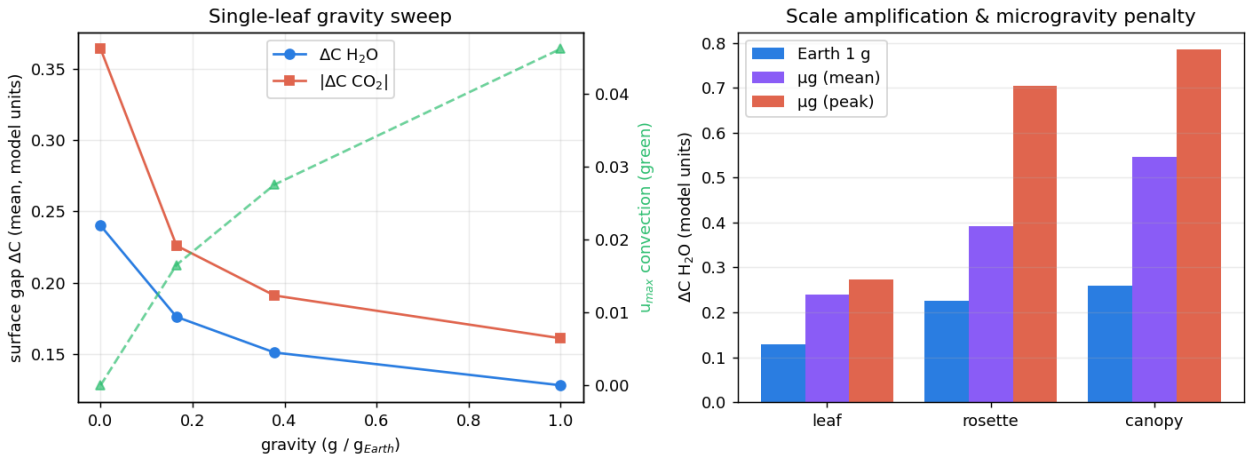


Figure 3: **Gravity scaling and architectural amplification of surface gas gradients. Left**, Single-leaf gravity sweep from μg through Moon ($0.17\,g$) and Mars ($0.38\,g$) to Earth ($1\,g$): convective peak velocity $u_{\max}$ scales as $\propto \sqrt{g}$, while mean surface H_2O and $|CO_2|$ concentration gaps steepen monotonically as gravity falls. **Right**, Architectural amplification across isolated leaf, overlapping rosette, and microgreen canopy scales at $1\,g$ and μg (`F4_gravity_scale.png`).

Table 2: **Steady-state lattice-Boltzmann boundary-layer diagnostics, physical surface CO_2/O_2 perturbations, and closed-loop spatial assimilation across gravity and plant scales.** Compiled from `T2_model_sweep.csv`, `T5_physical_prediction.csv`, `T12_feedback_spatial.csv`, and `T13_boundary_layer.csv`. Every scenario converged to 0.00% drift in g_{bl} over the final 5 s of simulation.

Scenario	**Gravity**	$u_{\max}$ **(lat.)**	g_{bl} (mol m^{-2} s^{-1})	δ (mm)	Sh	ΔCO_2 **mean/peak** (ppm)	ΔO_2 **mean** (ppm)	**Closed-loop** A (%)
Leaf · Earth	$1.00\,g$	4.61×10^{-2}	0.997	0.67	22.4	3.8 / 5.9	2.7	98.9
Leaf · Mars	$0.38\,g$	2.75×10^{-2}	0.841	0.79	18.9	4.6 / 6.8	3.2	—
Leaf · Moon	$0.17\,g$	1.65×10^{-2}	0.712	0.93	16.0	5.4 / 7.8	3.8	—
Leaf · μg	$0.00\,g$	0.00	0.443	1.50	10.0	8.7 / 9.9	5.5	97.6
Rosette · Earth	$1.00\,g$	4.60×10^{-2}	0.540	1.23	16.4	7.1 / 17.1	4.8	—
Rosette · μg	$0.00\,g$	0.00	0.272	2.45	8.2	14.2 / 25.4	8.9	96.1
Canopy · Earth	$1.00\,g$	7.19×10^{-2}	0.133	5.02	1.5	8.7 / 15.3	5.7	—
Canopy · μg	$0.00\,g$	0.00	0.058	11.42	0.7	19.8 / 28.4	12.5	94.5

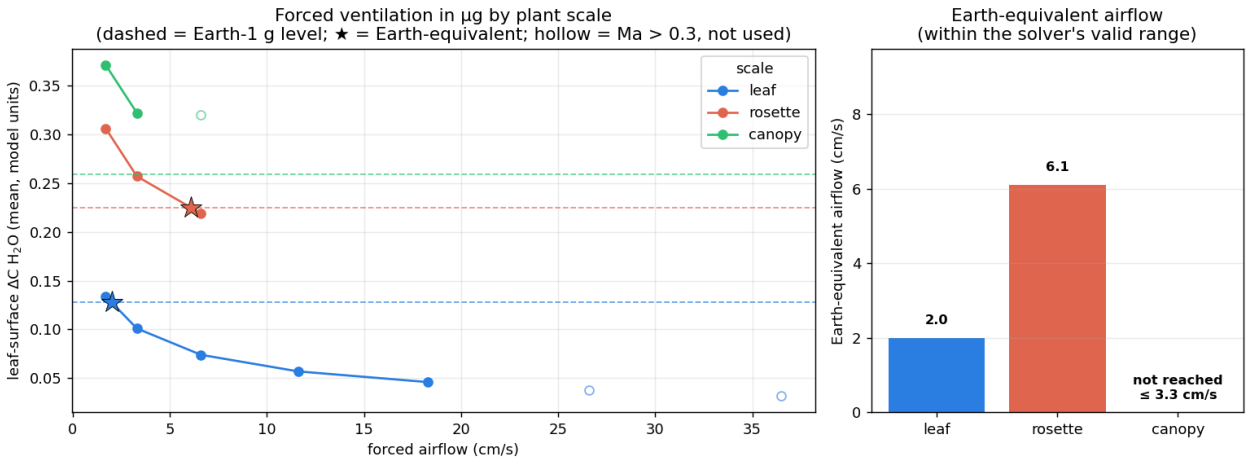


Figure 4: **Forced-ventilation requirement scales steeply with planting density. a**, Steady-state surface H_2O concentration gap (ΔC_{H_2O}, model units) versus forced inlet airspeed ($cm\,s^{-1}$) in microgravity across isolated leaf, rosette, and microgreen canopy scales (`F8_fan_by_scale.png`). Horizontal dashed lines indicate each architecture's $1\,g$ free-convection reference; hollow symbols exceed the lattice-Boltzmann low-Mach threshold (Ma > 0.3) and are excluded from crossing interpolation. **b**, Earth-equivalent forced airspeed required to restore $1\,g$ boundary-layer transport: $\approx 2.0\,cm\,s^{-1}$ for an isolated leaf and $\approx 6.1\,cm\,s^{-1}$ for a rosette (3× higher), whereas horizontal skimming flow does not restore the dense microgreen canopy within the low-Mach regime ($\leq 3.3\,cm\,s^{-1}$).

(14.2 ppm mean). In a dense microgreen canopy, boundary layers of adjacent shoots merge into a stand-wide stagnant layer: g_{bl} drops from $0.133\,mol\,m^{-2}\,s^{-1}$ at $1\,g$ to $0.058\,mol\,m^{-2}\,s^{-1}$ in μg ($\delta = 11.42$ mm), producing a mean surface CO_2 drawdown of 19.8 ppm (28.4 ppm peak) and a mean O_2 excess of 12.5 ppm. Replacing the reference elliptical cross-section with a realistic anatomical blade outline and transverse leaf curl (`NeuraLeaf` representation; Supplementary Table S4) further reduced g_{bl} by 3.8–5.6% (outline) and 16.8–18.6% (curl), with blade curl and microgravity compounding on the sheltered adaxial pocket to reach 2.6× the peak ΔCO_2 of a flat Earth blade. Three-grid Richardson extrapolation (0.288, 0.192, and 0.144 mm) confirmed monotonic convergence with a Celik et al. [20] fine-grid convergence index $GCI_{fine} = 3.16\%$ at $1\,g$ and 0.87% at $0\,g$ (Supplementary Table S3).

2.3. Forced ventilation reverses the microgravity penalty in a density-dependent manner

Because the microgravity boundary-layer penalty is purely transport-limited, it can be engineered away by replacing absent buoyant convection with forced airflow. Using a pressure-pinned outlet ($\rho = 1$) paired with a density-matched velocity inlet that delivers 99.6–100.9% of the set mass flux across all low-Mach runs (Ma ≤ 0.3; Supplementary Fig. S2b), we swept horizontal ventilation velocity across all three plant architectures in microgravity (Fig. 4; Supplementary Fig. S3).

For an isolated leaf in microgravity, a horizontal airflow of only $\approx 2.0\,cm\,s^{-1}$ ($\approx 2.1\,cm\,s^{-1}$ for CO_2) restores the surface concentration gap to its $1\,g$ free-convection value (Fig. 4a,b; `T6_forced_airflow.csv`). However, extending the sweep to denser architectures reveals a critical scaling law for hardware design: an overlapping rosette requires $\approx 6.1\,cm\,s^{-1}$ (3× higher airspeed) to achieve Earth parity, and even when VEGGIE-level airflow ($\sim 8\,cm\,s^{-1}$) restores the rosette's *mean* surface gap ($|\Delta C_{CO_2}| = 0.289$ vs. 0.298 at $1\,g$), it leaves highly uneven spatial ventilation across the sheltered crown (coefficient of variation CoV = 0.66 vs. 0.49 at $1\,g$; Supplementary Table S4). For the dense microgreen canopy, horizontal side-to-side airflow skims across the top of the shoot tips while inter-shoot air remains stagnant; its surface gap remains above the $1\,g$ level across the solver's valid low-Mach regime ($\leq 3.3\,cm\,s^{-1}$). Thus, a single ventilation speed tuned on sparse seedlings systematically under-ventilates a dense stand.

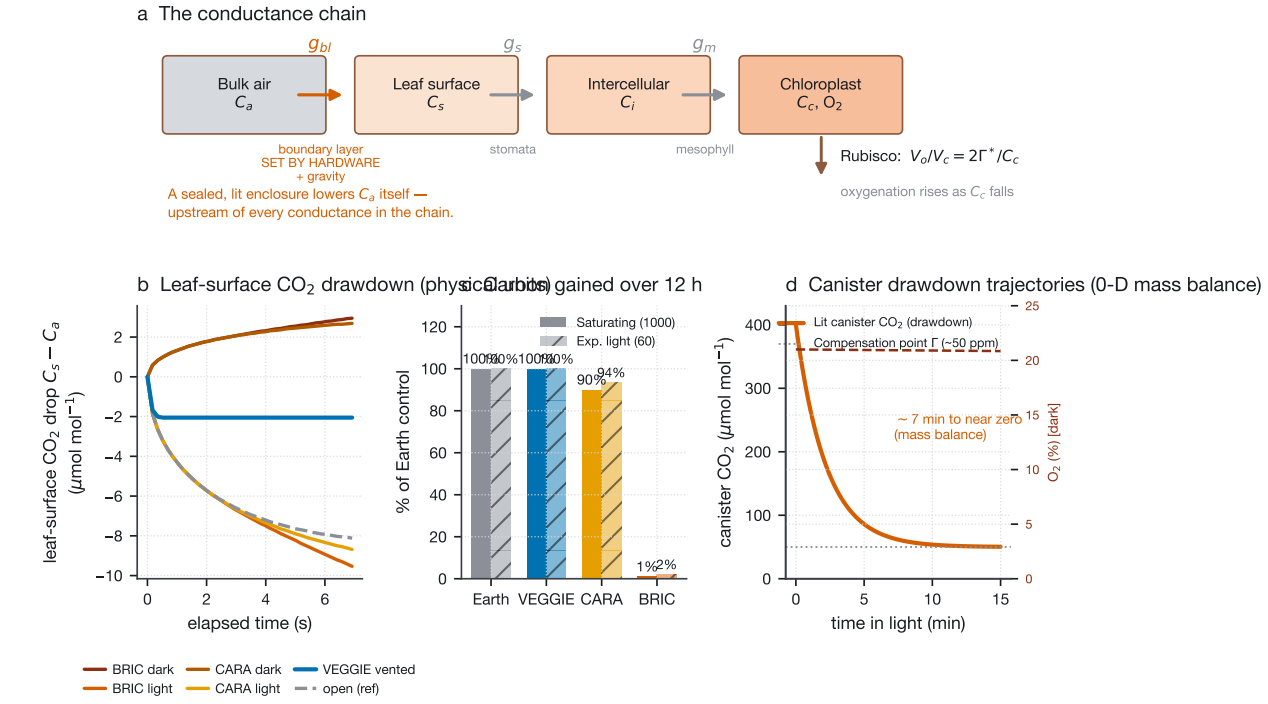


Figure 5: **Growth hardware, not gravity, sets the bulk atmosphere at the leaf.** **a**, Series conductance chain from cabin air across the enclosure seal, boundary layer (g_{bl}, the sole gravity-dependent step), stomata (g_s), and mesophyll (g_m) to Rubisco (C_c). **b**, Lattice-Boltzmann leaf-surface CO_2 drawdown ($C_s - C_a$, μmol mol^{-1}) over elapsed physical time (`T7_hardware_timeseries.csv`), showing identical surface diffusive penalties for BRIC and CARA in microgravity while VEGGIE forced airflow thins the boundary layer $\sim 4\times$. **c**, Integrated 12 h photoperiod carbon gain (% of unenclosed 1 g control) under experimental (60 μmol m^{-2} s^{-1}) and saturating (1000 μmol m^{-2} s^{-1}) irradiance (`T11_photosynthesis_feedback.csv`). **d**, Dynamic 0-D canister mass-balance trajectories (`T8_enclosure_timescales.csv`), showing exhaustion of headspace CO_2 to the compensation point within $\sim$ 7 min in illuminated sealed BRIC canisters and gradual O_2 depletion toward hypoxia ($\sim$ 6.5 d) in dark canisters (`fig01_hardware_atmosphere.pdf`).

2.4. Spaceflight growth hardware as boundary conditions: illuminated sealed canisters are CO_2-starvation chambers

Whereas gravity and fan speed govern the sub-millimetre boundary layer ($C_a \to C_s$), the growth container governs the bulk enclosure atmosphere ($C_{\text{cabin}} \to C_a$). We modelled the three primary ISS plant-growth hardware classes both as lattice-Boltzmann boundary conditions (zero-flux walls for hermetically sealed BRIC, semi-permeable relaxing walls for micropore-taped CARA, and forced inlet/outlet channel flow for VEGGIE; Supplementary Fig. S4) and as coupled zero-dimensional (0-D) headspace mass balances over a 12 h photoperiod (Fig. 5; Supplementary Table S1).

The two scales of transport separate cleanly across the three hardware classes:

- **BRIC (hermetically sealed)**: With zero gas exchange across the canister wall, plant metabolism drives runaway drift of the small headspace (30 mL air over 3 cm^2 leaf area; Fig. 5d). In the dark, respiratory CO_2 reaches 1% (10,000 ppm) in $\approx$ 9 h and O_2 falls from 21% to hypoxic 5% in $\approx$ 6.5 d. In the light, the $\sim$ 400 ppm of headspace CO_2 ($\approx$ 0.5 μmol total carbon) is fixed by photosynthesis in $\approx$ 7 min. Once coupled to CO_2-limited assimilation, headspace CO_2 collapses to the compensation point ($\Gamma \approx 50$ ppm) within minutes and remains pinned there for the rest of the photoperiod, reducing 12 h cumulative carbon gain to 1% of an unenclosed Earth

control (Fig. 5c). Sensitivity sweeps across headspace volume (15–60 mL), leaf area (1–10 cm^2), irradiance (60–1000 $\mu mol\,m^{-2}\,s^{-1}$), and cabin CO_2 (400–1200 ppm) confirm that rapid CO_2 starvation is an invariant property of sealed illuminated canisters (0.13–11.7% 12 h carbon gain across the entire parameter envelope; Supplementary Table S7).

- **CARA (gas-permeable micropore surgical tape)**: Diffusive exchange across the taped lid and perimeter (effective permeance 0.006 $mol\,m^{-2}\,s^{-1}$) resupplies CO_2 from the spacecraft cabin, sustaining ≈ 90–95% of Earth 12 h carbon gain (Fig. 5c; Supplementary Table S7). Crucially, however, passive tape permeance does not generate convective flow inside the dish: in microgravity, the leaf-surface boundary-layer gradient inside CARA is indistinguishable from that of a sealed BRIC dish (Fig. 5b; Supplementary Fig. S4). Thus, CARA solves the *enclosure-atmosphere* problem while leaving the *microgravity boundary-layer* problem intact.

- **VEGGIE (actively ventilated)**: Forced air circulation simultaneously pins the bulk atmosphere to cabin CO_2 and thins the leaf boundary layer, cutting the surface concentration gap $\sim 4\times$ relative to unventilated dishes (Fig. 5b) and sustaining ≈ 100% of Earth 12 h carbon gain (Fig. 5c).

Simulations inside exact two-dimensional cross-sections of the flight Petri dishes (60 mm round BRIC with 7.4 mm headspace above agar; 120 mm square CARA with 9.8 mm headspace; Supplementary Figs. S6–S7 and Table S6) revealed an additional geometric constraint: because the shallow 7–10 mm dish headspace suppresses vertical convective plumes even at 1 g, g_{bl} inside a shallow Petri dish at 1 g is already reduced to ≈ 0.25 $mol\,m^{-2}\,s^{-1}$, narrowing the 1 $g/\mu g$ ratio inside the dish to 1.25–1.8×. Moreover, when transpiration is coupled to headspace humidity saturation inside a sealed BRIC dish (`T20`–`T22`), the headspace saturates within 1–2 min, shutting down transpirational buoyancy at 1 g so that the 1 g and 0 g boundary layers converge within minutes (Supplementary Fig. S7). Because enclosure CO_2 drawdown is a mass balance determined by leaf area and headspace volume, under symmetric thermal and sealing conditions **it occurs in the 1 g ground control just as rapidly as in spaceflight**. Consequently, canister carbon starvation cancels out of a standard flight-versus-ground differential contrast, but shifts the physiological baseline of both arms into severe carbon depletion.

2.5. Coupling CFD transport to Rubisco kinetics: enclosure drawdown dominates and converges the gravity gap

To quantify how boundary-layer conductance (g_{bl}) and enclosure CO_2 (C_a) jointly regulate photosynthetic metabolism, we coupled the steady-state CFD g_{bl} table (`T13_boundary_layer.csv`) to a Farquhar–von Caemmerer–Berry (FvCB) C_3 biochemical model [5] parameterized with *in vivo Arabidopsis* Rubisco and electron-transport kinetics [21, 22] (Supplementary Table S2), solving recursively for the supply-equals-demand chloroplastic CO_2 concentration C_c and propagating the CFD-predicted surface O_2 build-up (ΔO_2 = 2.7–12.5 ppm) into local Γ^* (Fig. 6, Table 3).

At ambient CO_2 (400 ppm) and saturating light (1000 $\mu mol\,m^{-2}\,s^{-1}$), dropping from 1 g to μg lowers chloroplastic C_c from 238.9 to 227.0 ppm in an isolated leaf (A: 17.25 → 16.34 $\mu mol\,m^{-2}\,s^{-1}$, a 5.3% deficit), from 230.7 to 215.0 ppm in a rosette (A: 16.62 → 15.40 $\mu mol\,m^{-2}\,s^{-1}$, a 7.3% deficit), and from 189.2 to 149.3 ppm in a dense canopy (A: 13.30 → 9.80 $\mu mol\,m^{-2}\,s^{-1}$, a 26.3% deficit; φ rising from 31.1% to 36.4%). At the lower irradiance used in BRIC-LED flight OSD-522 (Q = 60 $\mu mol\,m^{-2}\,s^{-1}$), RuBP regeneration (A_j) limits assimilation and the microgravity deficit at ambient CO_2 is smaller: < 3% at leaf and rosette scale and up to 11% in a dense canopy (Fig. 6c).

By contrast, drawing the canister headspace down from 400 ppm to 100 ppm in a sealed lit BRIC canister shifts the leaf dramatically along its CO_2 response curve: at Q = 60 $\mu mol\,m^{-2}\,s^{-1}$, the

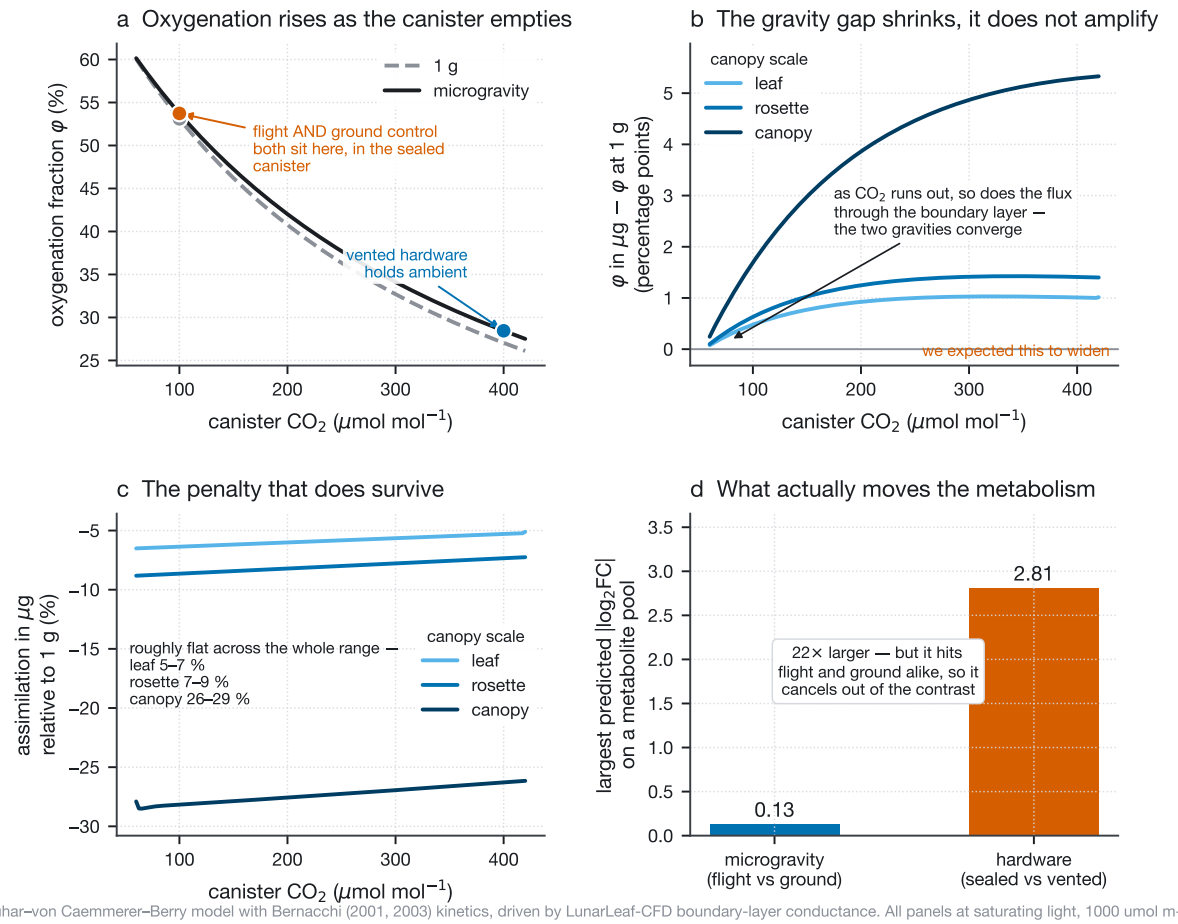


Figure 6: **Coupled CFD–FvCB Rubisco operating points: enclosure drawdown dominates and converges the microgravity boundary-layer gap.** **a**, Rubisco oxygenation fraction $\varphi = V_o/(V_o + V_c)$ as a function of canister CO_2 (C_a) at experimental irradiance ($60\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$, OSD-522; solid curves) and saturating irradiance ($1000\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$; dotted curves) for $1\,g$ (blue) and μg (vermilion) rosette conductances (`fig02_fvcb_operating_points.pdf`). **b**, The microgravity-minus-ground difference in oxygenation ($\Delta\varphi_{\mu g - 1g}$) shrinks toward zero as canister CO_2 is drawn down toward the compensation point. **c**, Relative microgravity assimilation deficit $(A_{1g} - A_{\mu g})/A_{1g}$ across leaf, rosette, and canopy scales at $Q = 60$ and $1000\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ (`T13_light_sensitivity.tsv`). **d**, Magnitude comparison of predicted $\log_2$ metabolite pool shifts: the sealed-versus-ventilated hardware effect is 22× larger than the flight-versus-ground microgravity effect.

Table 3: **Modelled CFD–FvCB operating points at experimental ($60\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$) and saturating ($1000\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$) irradiance.** Shown for rosette architecture ($g_{bl} = 0.540\,\mathrm{mol\,m^{-2}\,s^{-1}}$ at $1\,g$; $0.272\,\mathrm{mol\,m^{-2}\,s^{-1}}$ in μg). C_a and C_c in $\mu\mathrm{mol\,mol^{-1}}$; net assimilation A and photorespiration $R_p = 0.5\,V_o$ in $\mu\mathrm{mol\,m^{-2}\,s^{-1}}$; oxygenation fraction $\varphi = V_o/(V_o + V_c)$ in % (`T04_photorespiration_vs_gravity.csv`, `T05_operating_points.tsv`, `T13_light_sensitivity.tsv`).

Experimental arm	Irradiance Q ($\mu\mathrm{mol\,m^{-2}\,s^{-1}}$)	g_{bl} ($\mathrm{mol\,m^{-2}\,s^{-1}}$)	C_a (ppm)	C_c (ppm)	φ (%)	A ($\mu\mathrm{mol\,m^{-2}\,s^{-1}}$)	R_p ($\mu\mathrm{mol\,m^{-2}\,s^{-1}}$)
Ventilated control (μg)	60 (BRIC-LED)	0.272	400	334.6	18.0	4.90	0.68
Sealed ground ($1\,g$)	60 (BRIC-LED)	0.540	100	88.2	45.8	0.94	0.97
Sealed flight (μg)	60 (BRIC-LED)	0.272	100	87.2	46.2	0.88	0.94
Ventilated Earth ($1\,g$)	1000 (saturating)	0.540	400	230.7	27.0	16.62	4.09
Ventilated flight (μg)	1000 (saturating)	0.272	400	215.0	28.5	15.40	4.16
Sealed ground ($1\,g$)	1000 (saturating)	0.540	100	75.5	53.1	2.40	2.12
Sealed flight (μg)	1000 (saturating)	0.272	100	73.6	53.7	2.20	2.03

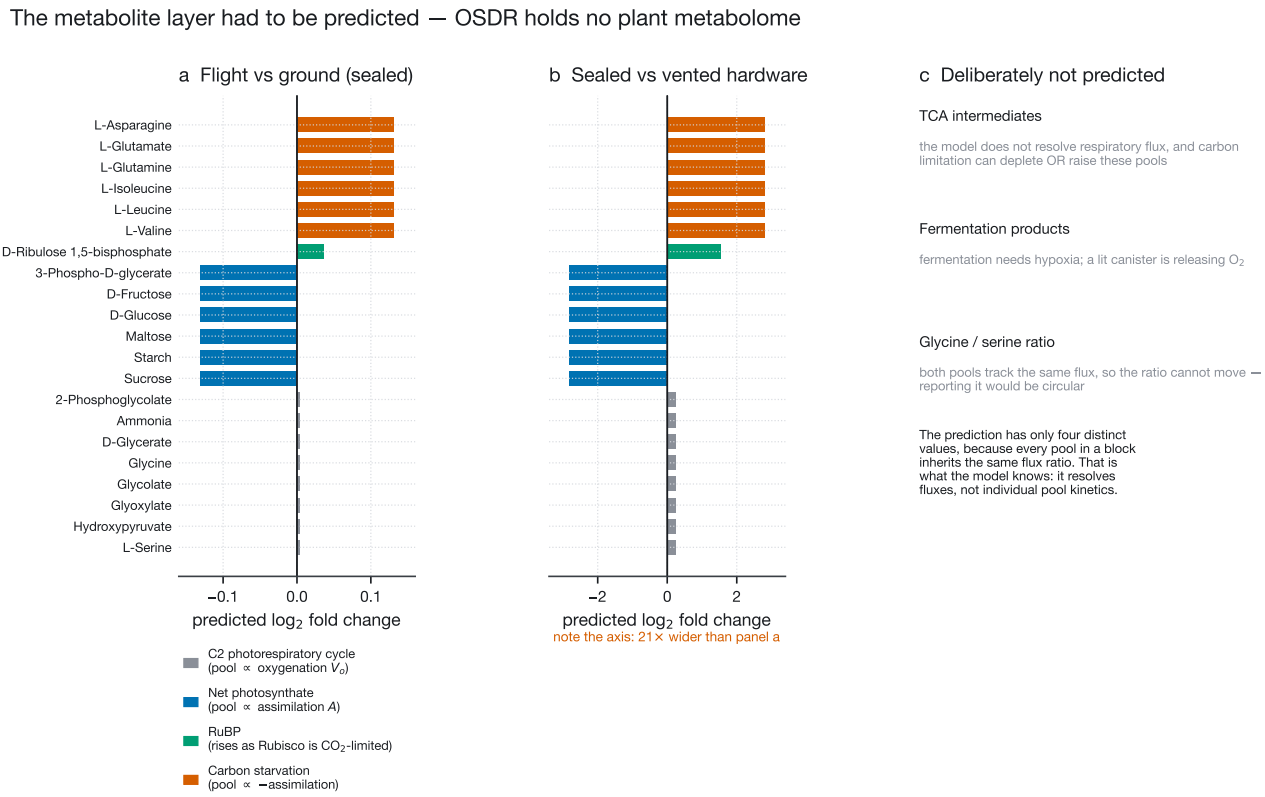


Figure 7: **The 21-compound predicted metabolite hypothesis scaffold.** Quasi-steady-state flux-propagated $\log_2$ fold-change predictions ($\log_2(P_{\mathrm{arm1}}/P_{\mathrm{arm2}}) = \log_2(J_{\mathrm{arm1}}/J_{\mathrm{arm2}})$) across Tier 1 stoichiometric intermediates (driven by V_o and A) and Tier 2 carbon-starvation / RuBP pools (driven by $-C_c$ and $-A$), comparing the flight-versus-ground microgravity contrast against the sealed-versus-ventilated enclosure contrast (`T06_predicted_compounds.tsv`, `fig03_predicted_metabolome.pdf`). All values are explicitly labelled as model output.

Rubisco oxygenation fraction φ jumps from 18.0% **to** 46.2%, and net assimilation A drops more than five-fold from 4.90 to 0.88 μmol m^{-2} s^{-1} (53.7% oxygenation at $Q = 1000\,\mu$mol m^{-2} s^{-1}; Fig. 6a, Table 3). Crucially, this coupling refuted our own initial hypothesis. We had hypothesized that canister CO_2 drawdown would *amplify* the microgravity boundary-layer penalty. Instead, the $1\,g$ and μg curves **converge** as C_a approaches Γ^* (Fig. 6b): because the concentration drop across the boundary layer is $\Delta C_{bl} = A/g_{bl}$, as $A \to 0$ near the compensation point, the flux across the boundary layer vanishes and gravity has nothing left to impede. Across the 21 modelled metabolic pools, the sealed-versus-ventilated enclosure effect is 22× **larger** than the flight-versus-ground microgravity effect (Fig. 6d).

2.6. The OSDR plant metabolomics gap and a falsifiable 21-compound hypothesis scaffold

Multi-omics pathway integration across central carbon and photorespiratory metabolism requires metabolite measurements alongside transcriptomics and proteomics. We queried all 567 studies in the NASA Open Science Data Repository (`T03_osdr_metabolomics_census.tsv`; Supplementary Table S8) and identified 12 studies with metabolite-profiling assays: all 12 are mammalian (*Homo sapiens*, *Mus musculus*, *Rattus norvegicus*) or microbial. **No plant spaceflight study in NASA OSDR contains a metabolome.**

To enable trimodal pathway topology testing and provide a concrete target list for future chemical profiling of archived flight tissues, we formulated a transparent 21-compound predicted

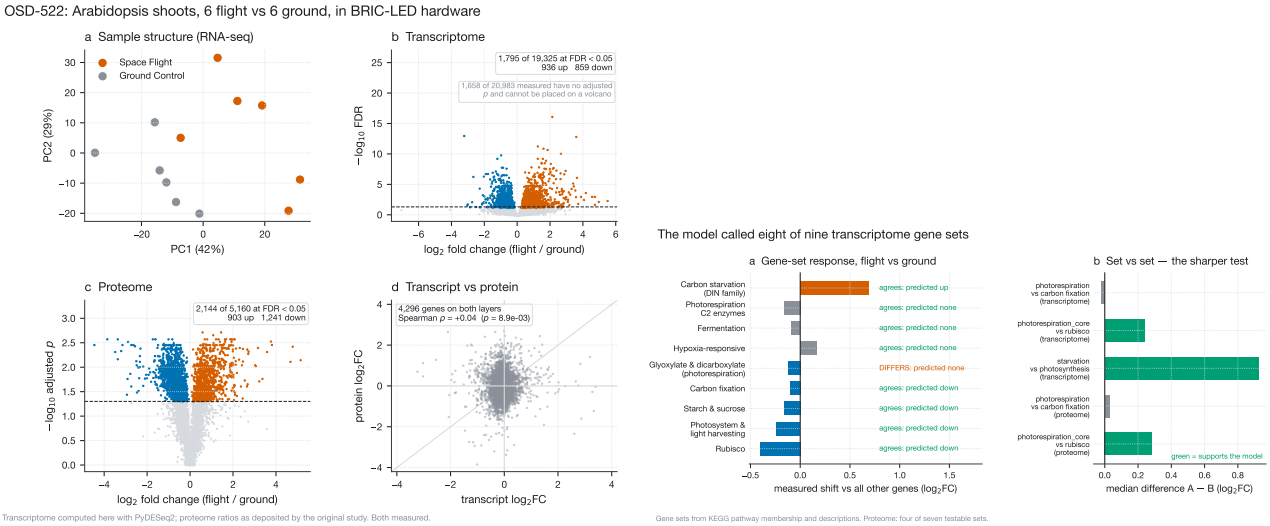


Figure 8: **OSD-522 (BRIC-LED) shoot multi-omics profiling and directional gene-set falsification test. Left** (`fig04_osd522_omics.pdf`): **a**, Principal component analysis of OSD-522 shoot RNA-seq libraries (6 flight vs. 6 ground). **b**, Transcriptome volcano plot (1,795 DEGs at FDR < 0.05). **c**, Shoot proteome volcano plot (5,160 proteins). **d**, Transcript-versus-protein $\log_2$ FC comparison across 4,296 co-measured loci. **Right** (`fig05_blind_test.pdf`): **a**, Directional predictions versus observed Mann–Whitney median $\log_2$ FC shifts across nine KEGG-derived gene sets in the transcriptome (8 of 9 matched) and proteome (4 of 7 matched). **b**, Set-versus-set relative contrasts demonstrating selective maintenance of core photorespiratory enzymes above Rubisco (+0.239 transcriptome, $p = 6.8 \times 10^{-3}$; +0.285 proteome, $p = 4.9 \times 10^{-2}$) and strong induction of *DIN* carbon-starvation markers over photosynthetic machinery (+0.923, $p = 5.3 \times 10^{-6}$).

metabolite layer (Fig. 7; Supplementary Table S9) by propagating the FvCB fluxes under quasi-steady state ($\log_2(P_1/P_2) = \log_2(J_1/J_2)$): eight C_2 photorespiratory intermediates driven by V_o (2-phosphoglycolate, glycolate, glyoxylate, glycine, L-serine, hydroxypyruvate, D-glycerate, ammonia), six photosynthetic carbon products driven by A (3-phosphoglycerate, sucrose, D-glucose, D-fructose, maltose, starch), RuBP driven by $-C_c$, and six amino-acid carbon-starvation markers driven by $-A$ (L-asparagine, branched-chain amino acids L-valine, L-leucine, L-isoleucine, and nitrogen-remobilization hubs L-glutamine and L-glutamate). Every table, figure, and interactive chart in the repository explicitly labels this layer as model prediction rather than measurement.

2.7. Testing against OSD-522 (BRIC-LED): the coupled model calls eight of nine transcriptome gene sets while ER unfolded protein response dominates globally

We tested the model's predictions against NASA OSDR study OSD-522 (BRIC-LED-001) [6], which grew *Arabidopsis thaliana* Col-0 shoots for 11 d inside sealed BRIC-LED canisters under 60 μmol m^{-2} s^{-1} illumination ($n = 6$ flight vs. $n = 6$ ground biological replicates) and provides paired RNA-seq and TMT quantitative proteomics from the same experiment (Fig. 8). Differential expression analysis of the shoot RNA-seq counts with PyDESeq2 [23] (`T01_osd522_transcriptome.tsv`) yielded 1,795 differentially expressed genes (DEGs) at Benjamini–Hochberg FDR < 0.05 out of 19,325 tested loci (1,658 low-count loci received NA adjusted p-values via standard DESeq2 independent filtering; Fig. 8a,b). Processing the deposited soluble and membrane shoot proteomics (`T02_osd522_proteome.tsv`) yielded ratios for 5,160 proteins (4,296 co-measured with RNA-seq; Fig. 8c,d).

Scoring the transcriptome and proteome against nine pre-specified KEGG and literature gene sets (`scripts/genesets.py`, `T07_falsification_osd522.tsv`; Fig. 8 right, Table 4) confirmed **eight of nine directional calls in the transcriptome** and **four of seven in the proteome**:

Table 4: **OSD-522 (BRIC-LED) multi-omics falsification scorecard across KEGG-derived gene sets and set-versus-set contrasts.** Mann–Whitney U shifts (gene-set median minus genome-wide background median) and two-sided p-values across the shoot transcriptome (19,325 loci) and shoot proteome (5,160 proteins) from `T07_falsification_osd522.tsv`.

Gene set / Contrast	Layer	n	Median $\log_2$ FC	Shift vs. bg	p-value	Predicted	Verdict
A. Single gene-set directional tests (vs. genome background)							
Photosynthesis apparatus	Transcriptome	31	−0.215	−0.238	3.8×10^{-6}	Down	Match
Rubisco subunits	Transcriptome	5	−0.381	−0.403	4.5×10^{-3}	Down	Match
Carbon fixation (`ath00710`)	Transcriptome	66	−0.082	−0.105	1.3×10^{-4}	Down	Match
Starch & sucrose (`ath00500`)	Transcriptome	142	−0.142	−0.165	3.5×10^{-6}	Down	Match
Carbon starvation (*DIN*)	Transcriptome	5	+0.708	+0.685	2.6×10^{-3}	Up	Match
Photorespiration core (C_2)	Transcriptome	13	−0.142	−0.164	0.092	None	Match
Glyoxylate/dicarboxylate (`ath00630`)	Transcriptome	76	−0.102	−0.125	3.2×10^{-4}	None	Mismatch (Down)
Fermentation	Transcriptome	6	−0.067	−0.090	0.596	None	Match
Hypoxia responsive	Transcriptome	3	+0.191	+0.169	0.248	None	Match
Photosynthesis apparatus	Proteome	19	−0.920	−0.820	5.0×10^{-4}	Down	Match
Rubisco subunits	Proteome	3	−0.490	−0.390	0.062	Down	Mismatch ($p = 0.06$)
Carbon fixation (`ath00710`)	Proteome	55	+0.050	+0.150	0.194	Down	Mismatch
Starch & sucrose (`ath00500`)	Proteome	68	−0.195	−0.095	0.133	Down	Mismatch
Photorespiration core (C_2)	Proteome	10	−0.205	−0.105	0.414	None	Match
Glyoxylate/dicarboxylate (`ath00630`)	Proteome	55	+0.080	+0.190	0.296	None	Match
Fermentation	Proteome	3	+0.510	+0.610	0.067	None	Match
B. Set-versus-set relative contrasts (immune to global plastid depression)							
Photorespiration core vs. Rubisco	Transcriptome	13 vs. 5	−0.142 vs. −0.381	+0.239	6.8×10^{-3}	A > B	Supports
Photorespiration core vs. Rubisco	Proteome	10 vs. 3	−0.205 vs. −0.490	+0.285	4.9×10^{-2}	A > B	Supports
Starvation (*DIN*) vs. Photosynthesis	Transcriptome	5 vs. 31	+0.708 vs. −0.215	+0.923	5.3×10^{-6}	A > B	Supports

1. **Photosynthetic apparatus repression**: Light-harvesting and photosystem subunits (`ath00195`/`ath00196`) fell significantly in both the transcriptome (median shift −0.238, $p = 3.8 \times 10^{-6}$) and proteome (−0.820, $p = 5.0 \times 10^{-4}$).
2. **Rubisco and Calvin-cycle repression**: Rubisco subunits (`RBCS1A-3B`, `RBCL`) fell strongly in the transcriptome (−0.403, $p = 4.5 \times 10^{-3}$; proteome −0.390, $p = 0.062$), alongside broader photosynthetic carbon fixation (`ath00710`; transcriptome −0.105, $p = 1.3 \times 10^{-4}$) and starch/sucrose metabolism (`ath00500`; transcriptome −0.165, $p = 3.5 \times 10^{-6}$).
3. **Carbon-starvation induction**: Classic dark-inducible / carbon-starvation markers (*DIN6/ASN1*, *DIN1*, *DIN10*, *BCAT2*) rose sharply in the transcriptome (+0.685, $p = 2.6 \times 10^{-3}$).
4. **Absence of bulk photorespiratory or hypoxic induction, but selective maintenance of C_2 enzymes over Rubisco**: Because canister CO_2 starvation occurs in both flight and ground canisters (and gravity converges at low CO_2), the model predicted no strong net induction of core photorespiratory enzymes (`photorespiration_core`: $p = 0.092$ transcriptome, $p = 0.414$ proteome; `MATCH`), fermentation ($p = 0.596$ transcriptome, $p = 0.067$ proteome; `MATCH`), or core hypoxia markers ($p = 0.248$ transcriptome; `MATCH`). Crucially, however, in set-versus-set contrasts that control for global transcriptional depression of plastid genes, core photorespiratory enzymes sat significantly **above Rubisco** in both the transcriptome (+0.239 $\log_2$ FC, $p = 6.8 \times 10^{-3}$) and the proteome (+0.285 $\log_2$ FC, $p = 4.9 \times 10^{-2}$; Table 4B), and carbon-starvation markers sat +0.923 $\log_2$ FC above the photosynthetic apparatus ($p = 5.3 \times 10^{-6}$).

To place these targeted carbon-metabolism shifts in the context of the whole cell, we integrated the OSD-522 transcriptome, proteome, and metabolite scaffold across 232 KEGG and MapMan pathways in PaintOmics 4 [24] (`T08`–`T10`; Fig. 9; Supplementary Fig. S8). Thirty-two pathways reached uncorrected combined Fisher $p < 0.05$ and six survived Benjamini–Hochberg FDR correction. The strongest signal in the dataset is one that gas transport does not explain: **protein processing in the endoplasmic reticulum** (the unfolded protein response, UPR; combined $p = 8.6 \times 10^{-9}$, FDR = 2.0×10^{-6}). Photosynthesis (`ath00195`, $p = 0.008$) and antenna proteins (`ath00196`, $p = 0.022$) were

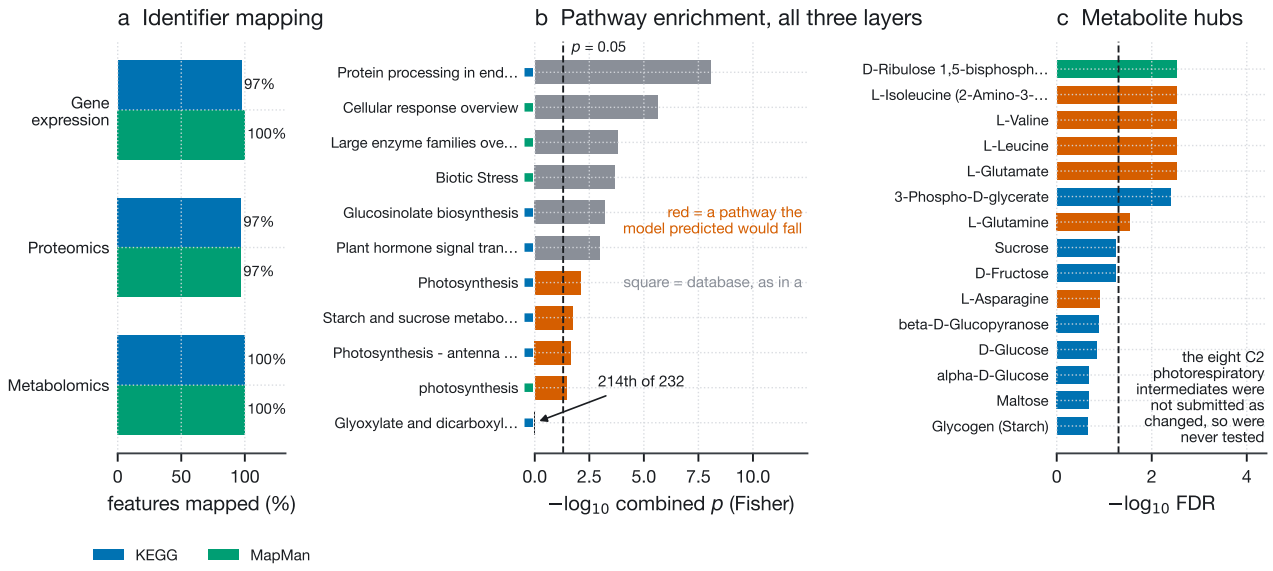


Figure 9: **PaintOmics 4 multi-omics pathway integration and empirical metabolite hub neighbourhood test.** **a**, Feature coverage across KEGG and MapMan databases. **b**, Combined pathway enrichment ranking across 232 pathways (`T08_paintomics_significant.tsv`, `T09_paintomics_carbon.tsv`): endoplasmic reticulum protein processing (UPR, $p = 8.6 \times 10^{-9}$) dominates globally; Photosynthesis ($p = 0.008$) and Antenna proteins ($p = 0.022$) reach uncorrected significance, while Glyoxylate and dicarboxylate metabolism (photorespiration) ranks 214th of 232 ($p = 0.994$, bottom 8%). **c**, Empirical metabolite hub analysis (`T10_metabolite_hubs.tsv`), testing each compound's immediate KEGG reaction neighbourhood for enrichment of measured OSD-522 DEGs: 7 of 15 hubs reach FDR < 0.05, exclusively in carbohydrate supply and amino-acid starvation blocks (`fig06_paintomics.pdf`).

enriched at uncorrected $p < 0.05$, whereas glyoxylate and dicarboxylate metabolism (photorespiration, `ath00630`) ranked **214th of 232 pathways** ($p = 0.994$, bottom 8%). An integration control omitting the predicted metabolite layer (`T17_integration_control_nometa.tsv`; Supplementary Table S10) and an independent pathway enrichment test outside PaintOmics (`T16_independent_pathways.tsv`; Supplementary Table S11) confirmed that the pathway rankings, the UPR dominance, and the photosynthesis/starch repression are completely invariant to the predicted metabolite layer. Meanwhile, ranking the predicted compounds by the empirical enrichment of measured DEGs in their immediate KEGG reaction neighbourhood (`T10_metabolite_hubs.tsv`; Fig. 9c) identified 7 of 15 compounds at FDR < 0.05, all carbon-supply or starvation hubs (sucrose, D-glucose, D-fructose, maltose, L-glutamine, L-glutamate, L-asparagine) and zero photorespiratory intermediates.

2.8. Cross-study evaluation across five OSDR spaceflight experiments: illumination separates all six flight contrasts

Finally, we tested whether the carbon-starvation and photorespiratory signatures scale across the ISS hardware ladder by assembling all available NASA OSDR *Arabidopsis* spaceflight studies that met strict pre-specified inclusion criteria: (i) green shoot, leaf, or whole-seedling tissue (excluding root-only studies because leaf photosynthetic gas transport makes no prediction for roots); (ii) RNA-seq with deposited GeneLab differential expression tables; and (iii) flight-versus-ground contrasts matched on genotype, ecotype, and illumination (`data/study_registry.tsv`; Table 5). Five studies yielding six matched contrasts met these criteria: sealed lit **OSD-522** (BRIC-LED, 60 μmol m^{-2} s^{-1}), sealed

Table 5: **NASA OSDR spaceflight studies evaluated in the cross-study hardware and illumination ladder.** Audit registry (`data/study_registry.tsv`) listing the five included studies (six matched flight-versus-ground contrasts) and six excluded studies with hardware class, tissue, genotype, mission duration, illumination, and inclusion/exclusion rationale.

Study	Hardware	Seal class	Tissue	Genotype	Duration	Light regime	Status / Reason
OSD-522	BRIC-LED	Sealed	Shoots	Col-0 WT	11 d	Lit ($60\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$)	Included
OSD-38	BRIC-PDFU	Sealed	Seedlings	Col-0 WT	8 d	Dark	Included
OSD-321	BRIC-PDFU	Sealed	Seedlings	Col-0 WT	14 d	Dark	Included
OSD-678 (lit)	CARA plate	Micropore tape	Leaves	Col-0 WT	12 d	Lit (4–$6\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$)	Included
OSD-678 (dark)	CARA plate	Micropore tape	Leaves	Col-0 WT	12 d	Dark	Included
OSD-427	VEGGIE (VPS)	Ventilated	Leaves	Col-0 WT	21 d	Lit ($120\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$)	Included
OSD-120	CARA plate	Micropore tape	Root tips	Col-0, Wass.	12 d	Lit / Dark	Excluded (roots only)
OSD-193	APH / APEX	Ventilated	Roots	Col-0, *sku6*	14 d	Lit	Excluded (roots only)
OSD-281	APH / APEX	Ventilated	Roots	WS, *sku5*	14 d	Lit	Excluded (roots only)
OSD-218	APH / APEX	Ventilated	Unstated	Col-0, WS	14 d	Lit	Excluded (tissue unstated)
OSD-16	APEX-TAGES	Ventilated	Leaf / stem	Col-0 WT	12 d	Lit	Excluded (no DGE table)
OSD-7	ABRS / TAGES	Ventilated	Whole plant	Col-0 WT	12 d	Lit	Excluded (microarray)

dark **OSD-38** (BRIC-20) and **OSD-321** (BRIC-22), micropore-taped **OSD-678** (CARA, matched Col-0 lit cabin-wall and dark arms), and actively ventilated lit **OSD-427** (APEX-04 / VEGGIE, $120\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$) [6, 8, 9, 12, 13].

Evaluating the ten gene sets and set-versus-set contrasts across all six studies (`T11_hardware_ladder.tsv`, `T12_ladder_contrasts.tsv`; Fig. 10 left) revealed two sharp empirical findings:

1. **Illumination completely separates the six spaceflight contrasts**: On the primary carbon-starvation-over-photosynthesis contrast ($DIN-$photosynthesis apparatus), **every illuminated contrast is positive** (+0.41 in VEGGIE OSD-427, +0.92 in BRIC-LED OSD-522, and +2.58 in lit CARA OSD-678) and **every dark contrast is negative** (−0.39 in dark BRIC OSD-38, −1.95 in dark BRIC OSD-321, and −3.22 in dark CARA OSD-678; descriptive one-sided Mann–Whitney $p = 0.05$; Fig. 10b). This separation is physically consistent with our central mechanism: photosynthetic CO_2 drawdown and boundary-layer CO_2 depletion require light; in darkness, respiration releases CO_2 rather than consuming it.
2. **The simple one-dimensional ranking across enclosure classes does not hold**: Among the three illuminated contrasts, sealed BRIC-LED (+0.92) sits above ventilated VEGGIE (+0.41) as predicted, but the dimly lit CARA plates in OSD-678 (+2.58, grown in only 4–$6\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ of ambient cabin light near the light compensation point, and with unrecorded plate-sealing tape geometry in OSDR) sit highest (Spearman $\rho = -0.50$, $p = 0.67$; Fig. 10c).

3. Discussion

3.1. Two distinct physical scales of gas transport in spaceflight plant biology

By uniting a validated 2-D lattice-Boltzmann fluid and mass-transport solver with a mechanistic FvCB Rubisco model and testing their combined predictions across NASA OSDR, this study resolves a longstanding ambiguity in spaceflight plant biology. Two distinct transport bottlenecks operate simultaneously in orbital plant experiments, but at different spatial scales and with different consequences for differential multi-omics:

1. **The microgravity boundary layer ($\sim$ 0.7–11 mm, $C_a \rightarrow C_s$)** is a genuine gravity effect. Eliminating buoyancy-driven convection reduces g_{bl} by 2.0–2.3× across single leaves, rosettes, and canopies (Table 2). At ambient CO_2 (400 ppm) and moderate-to-high irradiance, this boundary-layer thickening depresses chloroplastic C_c, elevates Rubisco oxygenation φ, and reduces net assimilation by 5–9% in leaves/rosettes and 26–29% in dense microgreen canopies (or

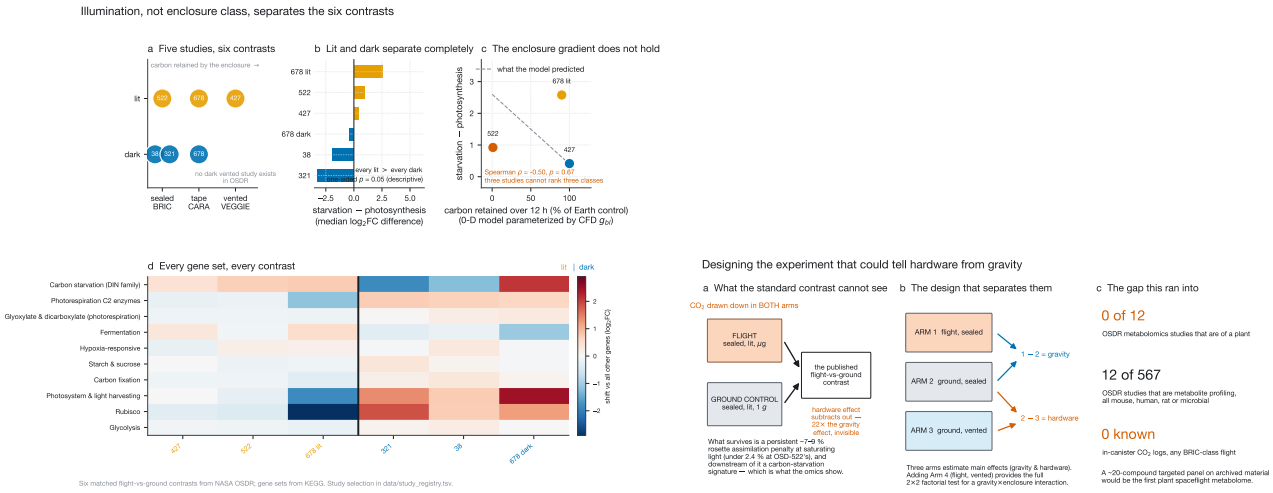


Figure 10: **Cross-study evaluation across six OSDR flight contrasts and experimental design to unconfound hardware from microgravity. Left** (`fig07_hardware_ladder.pdf`): **a**, Matrix of six flight-versus-ground contrasts across five OSDR studies. **b**, Illumination separates all three lit contrasts (+0.41, +0.92, +2.58) from all three dark contrasts (−0.39, −1.95, −3.22; one-sided Mann–Whitney $p = 0.05$) on the carbon-starvation (*DIN*) over photosynthesis contrast (`T12_ladder_contrasts.tsv`). **c**, Among lit studies, the predicted monotonic gradient across enclosure classes does not hold ($\rho = -0.50$, $p = 0.67$), with dimly lit CARA (`OSD-678`, 4–6 μmol m^{-2} s^{-1}) sitting highest. **d**, Heatmap of Mann–Whitney shifts across all ten evaluated gene sets (`T11_hardware_ladder.tsv`). **Right** (`fig08_implications.pdf`): **a**, Why standard two-arm spaceflight designs (sealed flight vs. sealed ground) cancel the 22× hardware starvation effect while measuring microgravity in carbon-starved plants. **b**, Proposed three-arm (adding vented ground) and 2 × 2 factorial (adding vented flight) designs that separate main effects of enclosure atmosphere, microgravity, and their interaction. **c**, Summary of the OSDR plant metabolomics and in-canister logging archive gaps.

< 3% and 11% at 60 μmol m^{-2} s^{-1}; Fig. 6c). Because this penalty is purely diffusive, horizontal forced ventilation of ≈ 2.0 cm s^{-1} (single leaf) to ≈ 6.1 cm s^{-1} (rosette) completely restores Earth-equivalent boundary-layer conductance (Fig. 4).

2. **Enclosure headspace drawdown** (10^1–10^2 mm, $C_{\mathbf{cabin}} \rightarrow C_a$) is a hardware effect. In an illuminated, hermetically sealed canister such as BRIC-LED, photosynthetic uptake exhausts the ~ 400 ppm headspace CO_2 to the compensation point in ~ 7 min, slashing 12 h carbon gain to $\approx 1\%$ of an unenclosed control and driving Rubisco oxygenation φ from 18% to 46% (Fig. 5, Fig. 6). This enclosure effect is $\approx 22\times$ larger than the microgravity boundary-layer effect, and because it is a volume mass balance, it occurs in the 1 g ground control canister just as rapidly as in flight.

This interaction explains why the microgravity boundary-layer penalty and canister carbon starvation cannot be treated as simple additive stresses. As canister CO_2 falls toward the compensation point, net assimilation A approaches zero, so the concentration drop across the boundary layer ($\Delta C_{bl} = A/g_{bl}$) vanishes and the 1 g and μg operating points converge (Fig. 6b). Likewise, inside a sealed BRIC Petri dish, transpirational humidification saturates the shallow 7.4 mm headspace within 1–2 min, extinguishing solutal buoyancy at 1 g so that even the physical conductances $g_{bl}(1\,g)$ and $g_{bl}(0\,g)$ converge (Supplementary Fig. S7). Consequently, a flight-versus-ground comparison in illuminated sealed hardware measures the residual effect of microgravity inside plants that their own growth canister has already severely carbon-starved (Fig. 10 right). Consistent with this transport-limited framework, whole-stand gas exchange of wheat in microgravity is indistinguishable from 1 g at saturating CO_2 [25]—the exact regime in which boundary-layer and enclosure CO_2 limitations are relieved.

3.2. What the model explains in the spaceflight multi-ome, and what it does not

In OSD-522 (BRIC-LED), the coupled model accurately predicted eight of nine directional gene-set shifts in the transcriptome and four of seven in the proteome, including concerted repression of photosystems, Rubisco, Calvin-cycle enzymes, and starch/sucrose synthesis, strong induction of *DIN6/ASN1* carbon-starvation markers, and selective preservation of photorespiratory enzymes relative to Rubisco (+0.239 $\log_2$ FC in RNA-seq, +0.285 in proteomics; Table 4). Across five independent OSDR experiments (six matched contrasts), illumination completely separated all lit contrasts from all dark contrasts on the starvation-over-photosynthesis axis (Fig. 10b).

Equally important are the three places where the data falsified or exceeded the gas-transport model:

1. **Dominance of the endoplasmic reticulum unfolded protein response (UPR)**: The single strongest pathway enrichment in OSD-522 is ER protein processing (`ath04141`, $p = 8.6\times10^{-9}$; Fig. 9b), a conserved spaceflight stress signature also highlighted in OSD-321 [13] that leaf gas transport does not explain. Biophysical and metabolic models of spaceflight adaptation must therefore treat ER proteostasis alongside gas transport rather than assuming a single master stressor.
2. **Absence of global photorespiratory pathway enrichment**: Although photorespiratory enzymes were maintained relative to repressed Rubisco, `ath00630` (glyoxylate and dicarboxylate metabolism) ranked 214th of 232 pathways in PaintOmics ($p = 0.994$), confirming that sealed-canister drawdown cancels between flight and ground arms rather than producing a large net flight-versus-ground photorespiratory induction.
3. **Irradiance and sealing confounders across the cross-study ladder**: Among illuminated studies, OSD-678 (CARA) showed the largest starvation-over-photosynthesis shift despite using micropore tape, likely because its lit arm grew in only 4–6 μmol m^{-2} s^{-1} of cabin light (near the photosynthetic light compensation point) and because exact tape area/permeance is not standardized in OSDR metadata.

3.3. Limitations and roadmap for spaceflight hardware and experimental design

Several limitations bound our conclusions. First, the lattice-Boltzmann solver is two-dimensional and employs the Boussinesq approximation ($\beta\Delta C \lesssim 0.5$) and low-Mach limit (Ma ≤ 0.3). While 1 g convective conductances are insensitive to chamber domain size ($\leq$ 7% change when domain dimensions are doubled; Supplementary Fig. S5 and Table S5), 0 g pure-diffusion conductances in 2-D depend logarithmically on the distance to the ambient boundary (g_{bl} dropping 33% for a leaf and 41% for a rosette when the box is doubled), making our open-box 1 $g/\mu g$ ratios conservative lower bounds and motivating full 3-D WebGPU simulations for dense canopies requiring turbulent through-flow. Second, wet-agar boundary conditions (`agarH2O = 1.3`) are numerically valid at 0 g ($g_{bl} = 0.185$–0.192 mol m^{-2} s^{-1}), but exceed the small-perturbation Boussinesq limit at 1 g ($\beta\Delta C \approx 1.3$) under both BGK and MRT collision schemes (`T23`, `T24`; Supplementary Table S6). Third, stomatal conductance ($g_s = 0.20$ mol m^{-2} s^{-1}) and mesophyll conductance ($g_m = 0.30$ mol m^{-2} s^{-1}) were held fixed at literature values rather than dynamically coupled to guard-cell CO_2/humidity signalling.

Three concrete experimental actions will resolve the hardware-versus-microgravity confound in future missions (Fig. 10 right):

1. **In-canister CO_2, O_2, and relative humidity logging**: Embedding miniature nondispersive infrared (NDIR) CO_2 and optical O_2 sensors inside BRIC and CARA hardware will convert our 0-D drawdown model into a direct measurement and render the archival BRIC transcriptome record quantitatively re-interpretable.

2. **A targeted 20-compound spaceflight plant metabolomics panel**: Profiling the 21 compounds in Table S9 (phosphoglycolate, glycolate, glycine, serine, 3-PGA, RuBP, sucrose, hexoses, maltose, Asn, BCAAs, Gln, Glu) on archived flight tissues will fill the complete absence of plant metabolomes in NASA OSDR (`T03`).
3. **Three-arm or 2×2 factorial ground controls**: Adding a ventilated ground control arm (or flying paired sealed and ventilated hardware in a 2×2 Gravity $\times$ Enclosure factorial design; Fig. 10b right) separates the 22× enclosure starvation main effect from the true microgravity main effect and their interaction.

4. Materials and Methods

4.1. Two-dimensional lattice-Boltzmann fluid and multi-species transport solver

Incompressible fluid flow was solved on a uniform two-dimensional Cartesian lattice ($\Delta x = 0.288\,\mathrm{mm}$, $\Delta t = 0.173\,\mathrm{ms}$) using the D2Q9 Bhatnagar–Gross–Krook (BGK) lattice-Boltzmann equation (`src/solver/lbm/LBMFluid.ts`):

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) - f_i(\mathbf{x}, t) = -\frac{1}{\tau_f}\left[f_i(\mathbf{x}, t) - f_i^{\mathrm{eq}}(\rho, \mathbf{u})\right] + \Delta t\, F_i(\mathbf{x}, t), \tag{3}$$

where $\mathbf{c}_i$ ($i = 0, \ldots, 8$) are the discrete D2Q9 velocity vectors, $c_s = 1/\sqrt{3}$ is the lattice speed of sound, $\nu = c_s^2(\tau_f - \frac{1}{2}) = 0.02$ is the kinematic viscosity in lattice units, and F_i is the discrete body-force term implemented via the exact second-order Guo forcing scheme [17]. An opt-in multiple-relaxation-time (MRT) collision operator in moment space [16] (`collision: 'mrt'`) was also implemented and verified across all four benchmark gates (`validation/mrt_check.ts`). Solid leaf and chamber wall nodes enforce no-slip velocity boundary conditions via halfway bounce-back.

Three gas species ($k \in \{\mathrm{CO_2}, \mathrm{O_2}, \mathrm{H_2O}\}$, stored as concentration excess over cabin ambient $c_k = C_k - C_{k,0}$) were transported simultaneously on D2Q5 advection–diffusion lattices (`src/solver/lbm/ScalarField.ts`) sharing the macroscopic fluid velocity $\mathbf{u}$:

$$g_{k,j}(\mathbf{x} + \mathbf{e}_j \Delta t, t + \Delta t) - g_{k,j}(\mathbf{x}, t) = -\frac{1}{\tau_k}\left[g_{k,j}(\mathbf{x}, t) - g_{k,j}^{\mathrm{eq}}(c_k, \mathbf{u})\right] + \Delta t\, w_j S_k(\mathbf{x}, t), \tag{4}$$

with lattice diffusivities $D_k = c_{s,5}^2(\tau_k - \frac{1}{2})$ set to $0.033 : 0.042 : 0.050$ matching the physical molecular diffusivity ratio $D_{\mathrm{CO_2}} : D_{\mathrm{O_2}} : D_{\mathrm{H_2O}} = 1.6 : 2.0 : 2.4 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$ at 25 °C. Gravity couples to the flow via a solutal Boussinesq body force (`src/solver/lbm/buoyancy.ts`):

$$\mathbf{f}(\mathbf{x}, t) = \rho\, \mathbf{g} \sum_k \beta_k \left(c_k(\mathbf{x}, t) - c_{k,\mathrm{ref}}\right), \tag{5}$$

where $\mathbf{g} = (0, -g)$ is scaled from $1\,g$ ($9.81\,\mathrm{m\,s^{-2}}$) through Mars ($0.378\,g$), Moon ($0.165\,g$), and microgravity ($0\,g$), and $c_{k,\mathrm{ref}}$ is either cabin ambient (0, for open/vented scenes) or the instantaneous enclosure mean (`buoyancyRef: 'enclosureMean'`, preventing uniform background drift in sealed dishes from generating spurious body forces).

4.2. Plant geometries, spaceflight hardware boundary conditions, and closed-loop feedback

Three biological architectures were constructed on a 128×96 production lattice ($36.9 \times 27.6\,\mathrm{mm}$; refined up to 384×288 in `validation/resolution_ladder.ts` and expanded up to 256×192 in `validation/domain_size.ts`): (i) an isolated horizontal leaf blade (52×8 cells, 1.5 cm chord, plus anatomical outlines and curled transverse sections in `src/leaf/`); (ii) an overlapping eight-blade rosette (`rosetteGeometry`, characteristic length 70 cells); and (iii) a microgreen canopy of nine upright

shoots above an adiabatic soil floor (`canopyGeometry`, 0.3× per-leaf flux for self-shading). Fluid nodes adjacent to leaf solid cells receive stomatal source/sink fluxes ($S_{CO_2} < 0$, $S_{O_2} > 0$, $S_{H_2O} > 0$; reversed and scaled by $R/A = 0.31$ in the dark). Boundary-layer conductance g_{bl} (mol m^{-2} s^{-1}), film thickness $\delta = D_{CO_2}/g_{bl}$ (mm), and Sherwood number $\mathrm{Sh} = L_{\mathrm{char}}/\delta$ are derived directly from the simulated surface flux and mean surface-to-bulk concentration gap, anchored to $g_{bl}(1\,g, \mathrm{leaf}) = 1.0\,\mathrm{mol\,m^{-2}\,s^{-1}}$ (`T13_boundary_layer.csv`).

Hardware boundary conditions on the domain border represent: (i) open reference (anti-bounce-back Dirichlet $c_k = 0$); (ii) sealed BRIC (zero-flux bounce-back, `membraneK = 0`); (iii) micropore-taped CARA (semi-permeable relaxation $c_k \leftarrow (1 - k_{\mathrm{mem}})c_k$ on perimeter fluid cells each step, `membraneK = 0.01`); and (iv) ventilated VEGGIE (horizontal channel with density-matched equilibrium inlet at fan speed U and pressure-pinned outlet $\rho = 1$). When closed-loop CO_2 feedback is enabled (`co2Ambient = 8`), local surface uptake scales as $m_{CO_2}(c) = [(C_a + c - \Gamma)/(C_a + c - \Gamma + K_C)]/m_{\mathrm{amb}}$ with $\Gamma = 0.125\,C_a$ and $K_C = 0.75\,C_a$. When humidity-limited transpiration is enabled (`h2oSatExcess = 1.3`), local H_2O release scales as $\max(0, 1 - c_{H_2O}/c_{\mathrm{sat}})$.

4.3. Coupled 0-D enclosure mass balance and Farquhar–von Caemmerer–Berry (FvCB) model

Canister-scale CO_2 drawdown over a 12 h photoperiod (`validation/feedback_0d.py`, `scripts/enclosure_sensitivity.py`) integrates the ordinary differential equation for well-mixed headspace concentration $C_a(t)$ in volume $V = 30$ mL over leaf area $A_{\mathrm{leaf}} = 3\,\mathrm{cm^2}$ and dish seal area $A_{\mathrm{dish}} = 28\,\mathrm{cm^2}$:

$$\rho_{\mathrm{air}} V \frac{\mathrm{d}C_a}{\mathrm{d}t} = -A_{\mathrm{leaf}}\,A_{\mathrm{net}}(C_s(t)) + A_{\mathrm{dish}}\,k_{\mathrm{seal}}\,(C_{\mathrm{cabin}} - C_a(t))\,, \tag{6}$$

where $\rho_{\mathrm{air}} = 41.6\,\mathrm{mol\,m^{-3}}$ (25 °C), $k_{\mathrm{seal}} = 0$ (BRIC), $0.006\,\mathrm{mol\,m^{-2}\,s^{-1}}$ (CARA micropore tape), or ∞ (VEGGIE), and $C_s(t)$ is coupled to $C_a(t)$ through g_{bl}.

In `scripts/fvcb.py` (mirrored to 10^{-6} parity in `public/explorer/app/fvcb.js`), steady-state g_{bl} from `T13_boundary_layer.csv` is placed in series with stomatal conductance $g_s = 0.20\,\mathrm{mol\,m^{-2}\,s^{-1}}$ and mesophyll conductance $g_m = 0.30\,\mathrm{mol\,m^{-2}\,s^{-1}}$ to give total conductance to Rubisco:

$$g_{\mathrm{tot}} = \left(\frac{1}{g_{bl}} + \frac{1}{g_s} + \frac{1}{g_m}\right)^{-1}, \qquad C_c = C_a - \frac{A}{g_{\mathrm{tot}}}. \tag{7}$$

Biochemical demand $A(C_c) = \min(A_c, A_j) - R_d$ is solved simultaneously with diffusive supply (Eq. 7) by bisection over $C_c \in [\Gamma^*, C_a]$ using Bernacchi et al. [21, 22] temperature responses at 25 °C ($V_{c\,\max,25} = 90.0\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$, $J_{\max,25} = 153.0\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$, $R_{d,25} = 1.35\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$, $K_{c,25} = 404.9\,\mu\mathrm{mol\,mol^{-1}}$, $K_{o,25} = 278.4\,\mathrm{mmol\,mol^{-1}}$, $\Gamma^*_{25} = 42.75\,\mu\mathrm{mol\,mol^{-1}}$ at $O_{\mathrm{ref}} = 210\,\mathrm{mmol\,mol^{-1}}$; Supplementary Table S2). Local surface O_2 build-up from CFD shifts the compensation point as $\Gamma^*(T, O_{\mathrm{local}}) = \Gamma^*(T) \cdot (O_{\mathrm{local}}/O_{\mathrm{ref}})$. From the fixed-point solution (C_c, A), carboxylation rate $V_c = (A + R_d)/(1 - \Gamma^*/C_c)$, oxygenation rate $V_o = 2(\Gamma^*/C_c)V_c$, photorespiration $R_p = 0.5\,V_o$, and oxygenation fraction $\varphi = V_o/(V_o + V_c)$ are evaluated exactly.

4.4. NASA OSDR multi-omics processing, gene-set falsification, and PaintOmics 4 integration

Unnormalized RSEM gene-level counts for OSD-522 ($n = 6$ flight, $n = 6$ ground shoot libraries) were retrieved via the NASA OSDR API (`scripts/00_fetch_osdr.py`) and analyzed with PyDESeq2 [23] (`scripts/01_dge_rnaseq.py`) using design $\sim$ `condition` with Cook's distance outlier refitting and independent filtering (`T01_osd522_transcriptome.tsv`). Deposited TMT soluble and membrane shoot proteomics tables from OSD-522 were merged by UniProt/AGI accession (`scripts/02_prep_proteomics.py`, `T02_osd522_proteome.tsv`). Gene sets (`scripts/genesets.py`) were retrieved live from the KEGG REST API [26, 27] (`ath00630`, `ath00710`, `ath00500`, `ath00010`, `ath00195`, `ath00196`) alongside curated Araport11 locus sets

for core photorespiratory enzymes (`photorespiration_core`, $n = 13$), Rubisco ($n = 5$), *DIN* carbon-starvation markers ($n = 5$), fermentation ($n = 6$), and core hypoxia markers ($n = 3$). Directional falsification tests (`scripts/04_falsification_check.py`) used two-sided Mann–Whitney U tests of gene-set $\log_2$ FC distributions against the genome-wide background (`T07_falsification_osd522.tsv`). Multi-omics pathway enrichment was performed in PaintOmics 4 [24] across KEGG and MapMan ontologies (saved job record `oK2h66740Q`, verified against metabolite-free control `T17_integration_control_nometa.tsv` and external Fisher/Mann–Whitney tests `T16_independent_pathways.tsv`). MapMan pathway membership (`scripts/mapman.py`) was reconstructed deterministically from GoMapMan Araport11 bin assignments and vendored MapManStore diagram layouts (`data/mapman/PROVENANCE.md`). Cross-study evaluation (`scripts/07_hardware_ladder.py`) ingested GeneLab differential expression tables for OSD-522, OSD-38, OSD-321, OSD-678, and OSD-427 according to the audit rules in `data/study_registry.tsv`.

4.5. Use of AI coding and drafting assistants

In accordance with publisher guidelines on artificial intelligence transparency, the author discloses that AI coding and drafting assistants (Claude Code, Anthropic; Google Antigravity) were used under the author's direct supervision to implement solver and pipeline scripts, generate figures, and format manuscript drafts. The author independently verified all physical equations, numerical validation gates, OSDR data processing, statistical tests, and bibliographic citations, and takes full responsibility for the integrity of the work.

Data and Code Availability

All source code (TypeScript/React 2-D lattice-Boltzmann CFD workbench, headless validation suite, Python FvCB and multi-omics pipeline, and interactive web applications), derived tables (`T1`–`T24` and `T01`–`T17` with machine-verified provenance in `results/MANIFEST.tsv`), publication figures, and LaTeX/Word manuscript sources are openly available at https://github.com/dr-richard-barker/Spaceflight-Plant-Biophysics-Multiomics under the MIT (code) and CC-BY-4.0 (manuscript and figures) licenses, and archived on Zenodo. Primary spaceflight multi-omics datasets are publicly available in the NASA Open Science Data Repository [28] under accessions OSD-522 [6], OSD-38 [12], OSD-321 [13], OSD-678 [8], and OSD-427 [9]. Empirical whole-plant chamber gas-exchange data from Chew et al. [18] are archived at Edinburgh DataShare (https://doi.org/10.7488/ds/3453) [19].

CRediT Authorship Contribution Statement

Richard J. Barker: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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